Delta-Neutral Dynamic Hedging of a Short Index Put

Black-76, Smile Regimes, Heston & Deep Hedging — A TAIFEX Backtest

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Quantitative Strategy Interview | KGI Securities

April 2025 Backtest | TXO20000P5

Short 1 TXO 20,000 Put | Expiry 2025/04/16 | Hedged with TX Futures

Outline

Executive Summary

- **Task:** run a delta-neutral dynamic hedge of a **short** TAIFEX index put across the 19 trading days to expiry, executing only at daily settlement prices.
- **Twist:** the window contained the **April 2025 tariff shock** — a -2.7σ single-day jump, not a diffusion move.
- **Approach:** five hedging models of increasing sophistication, each under strict no-lookahead discipline, with full P&L attribution.

Headline result — net P&L vs. the Black-76 baseline (−NT$34,380)

Deep Hedging (+NT$2,946) > MV Delta (+NT$2,136) > Baseline > Heston (−NT$3,950)

Sophistication did *not* guarantee performance — **objective design and turnover control did**.

1. Mandate & Setup

The Mandate

Position (inception 2025/03/19)

- Short 1 `TXO20000P5`: monthly put, strike $K = 20{,}000$, expiry 2025/04/16
- Premium received: 68 pts $\times$ 50 $= NT\$ + 3{,}400$

Objective

- Hold delta ≈ 0 by trading TX index futures
- Rebalance daily at the **TAIFEX settlement price** (no intraday fills)

Why a short put needs a short hedge

A put has delta $-d$. Being *short* the put gives portfolio delta $+d$.

To neutralise, **sell** futures:

$$h = -d \times \frac{50}{200} = -d \times 0.25$$

TX contracts (fractional allowed).

Backtest window: 2025/03/19 $\rightarrow$ 2025/04/16 (19 trading days). Final settlement (SOQ) $= 19{,}548$.

Instruments & Conventions

Contract specifications

	Multiplier	Unit
TXO (option)	NT$50/pt	1 contract
TX (futures)	NT$200/pt	1 contract

Transaction costs (TAIFEX schedule)

- TX: exch. fee NT$20 + tax $0.00002 \times 200F$ (≈NT$108) per contract $\times |\Delta h|$
- TXO: exch. fee NT$10 + tax $0.001 \times$prem (≈NT$13) at inception
- Broker commission excluded; slippage 0

Modelling conventions

- **Black-76**: TX futures *is* the forward (TXO and TX share expiry ⇒ no carry adjustment)
- Day count: calendar days / 365
- Risk-free: CBC 31–90d CP rate ($\approx 1.6\%$)
- IV back-solved daily from same-day settlement (bisection)

Expiry handling

TX/TXO settle to 0 on 04/16 by convention ⇒ substitute official SOQ = 19,548; put intrinsic = 452.

2. Methodology & Discipline

Pricing & Greeks — Black-76

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

$$P = e^{-rT}[K\,N(-d_2) - F\,N(-d_1)]$$

$$\Delta_{\text{put}} = -\,e^{-rT} N(-d_1)$$

Daily loop: solve σ_t from $P_t \Rightarrow$ Greeks $\Rightarrow$ hedge $h_t = -|\Delta_t| \times 0.25 \Rightarrow$ P&L.

P&L attribution identity

$$\underbrace{\Theta}_{+} + \underbrace{\tfrac{1}{2}\Gamma(\Delta F)^2}_{-} + \underbrace{\mathcal{V}\Delta\sigma}_{\pm} + \underbrace{h\,\Delta F}_{\text{hedge}} + \text{resid}$$

Every day decomposed into theta / gamma / vega / delta + jump residual — they must reconcile to net P&L.

Backtest Discipline — No Lookahead

1. On date t, use only data in $[t_0, t]$ — never future prices or vol.
2. IV on date t back-solved from the **same-day** settlement only.
3. Hedge trades execute at date-t settlement, using date-t Greeks.
4. Costs charged on every day with a position change ($|\Delta h| > 0$).
5. Final settlement $=$ officially published TAIFEX SOQ $= 19{,}548$.

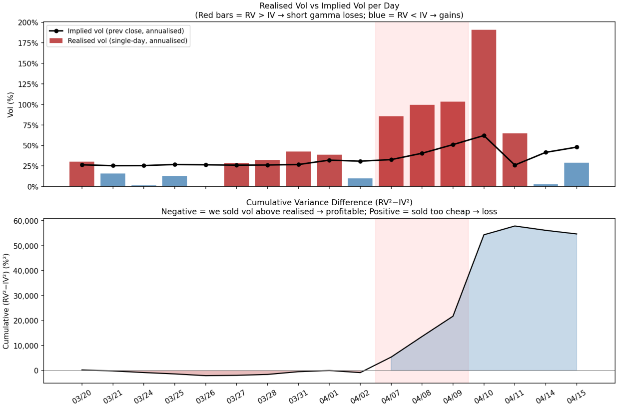
Why this matters

The whole exercise is only credible if the hedge at time t cannot "see" the crash before it happens. Each model — including the deep-learning one — is trained/calibrated strictly on information available *before* the decision.

3. The April 2025 Regime

A Jump, Not a Diffusion

- Crash–rally–crash: Apr 7 −10.5%, Apr 10 +9.9%, Apr 11 +2.9%.
- Implied vol spiked 26% → 62% in three sessions.
- Apr 7 return $\approx -2.7\sigma$ under log-normal Black-76 — a fat-tail event.



Consequence for the short gamma book

Realised vol far exceeded implied $\Rightarrow$ **gamma losses overwhelmed theta income**, and daily delta-hedging got whipsawed.

4. Models & Results

Five Hedging Models

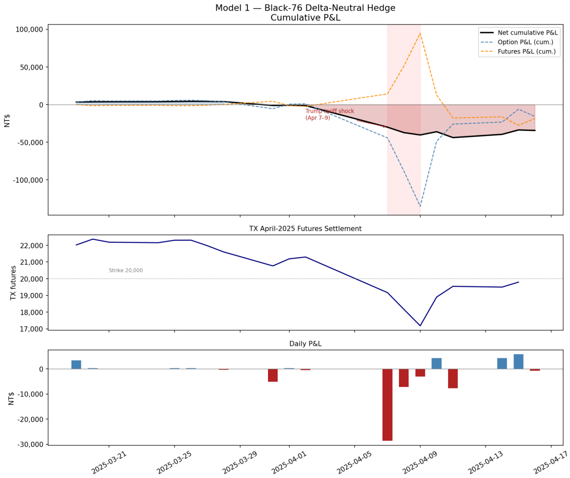
#	Model	Core idea
1	Black-76 Delta (baseline)	Textbook BS delta from same-day IV
2	Sticky-Strike vs Sticky-Delta	Which IV regime governs the hedge?
3	Minimum-Variance Delta	Adjust delta for the IV/spot correlation
4	Heston Stochastic Vol	Calibrate smile daily; model-implied delta
5	Deep Hedging (Buehler 2019)	LSTM policy minimising tail risk (CVaR)

Each adds realism over the last; the question is whether realism *pays* on a single jump-driven path.

Model 1 — Black-76 Baseline

P&L statement

Component	NT$
Premium received	+3,400
Option MTM	−19,200
Futures hedge P&L	−18,506
Transaction costs	−74
Net P&L	−34,380

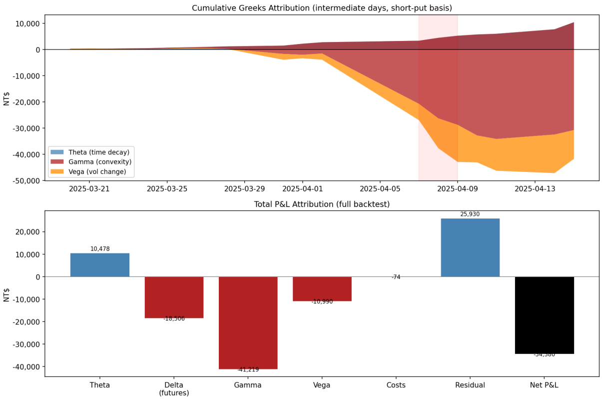


The puzzle

A "correctly" delta-hedged short put still lost NT$34,380. **Why?**

Model 1 — P&L Attribution Explains the Loss

Driver	NT$
Theta (decay)	+10,478
Gamma (convexity)	−41,219
Vega (vol MTM)	−10,990
Delta (futures)	−18,506
Residual (jump)	+25,930
Costs	−74
Net	−34,380



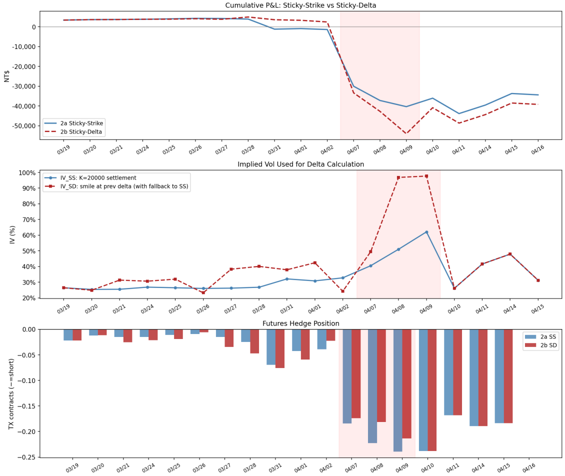
Reading

Gamma (−41k) dwarfed theta (+10k): the book was paid to decay but punished far more for the large moves. The big positive residual is the **jump** term — Black-76 cannot represent a -2.7σ day.

Model 2 — Sticky-Strike vs Sticky-Delta

Question: as spot moves, does the IV at our strike stay fixed (*sticky-strike*) or follow a fixed delta (*sticky-delta*)?

- 2a Sticky-Strike: IV from $K = 20{,}000$ each day $=$ Model 1.
- 2b Sticky-Delta: IV interpolated along the smile at the prior day's delta.
- Fallbacks near expiry / deep-ITM / unstable smile.



Result

Sticky-Delta net −NT$39,191 — *worse*. Its larger prior short lost more on the Apr 10 rally (whipsaw).

Model 3 — Minimum-Variance Delta

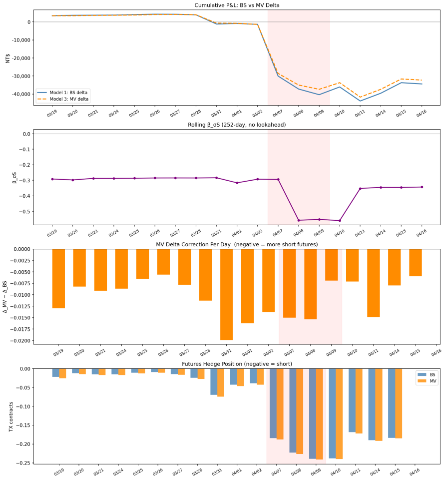
$$\Delta_{\text{MV}} = \Delta_{\text{BS}} + \mathcal{V}_{\text{BS}} \cdot \beta_{\sigma S}$$

$$\beta_{\sigma S} = \frac{\mathrm{Cov}(\Delta\sigma,\ \Delta S/S)}{\mathrm{Var}(\Delta S/S)}$$

- Captures the leverage effect: vol rises when spot falls $\Rightarrow$ extra short.
- β from rolling 60-day regression, initialised pre-t_0, updated daily (no lookahead).

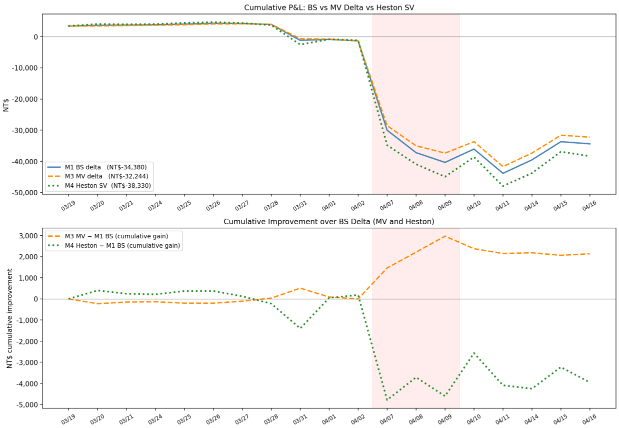
Result — best classical model

Net −NT$32,244 (+NT$2,136 vs. baseline). A small, **robust** improvement from a stable parameter.



Model 4 — Heston Stochastic Volatility

- Calibrate $\{\kappa, \theta, \sigma_v, \rho, v_0\}$ daily to the full put smile (86–177 strikes).
- Delta via finite difference on the calibrated model.
- Aims to capture smile dynamics BS delta misses.



Result — sophistication backfired

Net −NT\$38,330 (−NT\$3,950 vs. baseline).
Degenerate $v_0 \approx 0.015 \Rightarrow$ too-small delta $\Rightarrow$ under-hedged into the fall.
Estimation/calibration risk during crisis.

Model 5 — Deep Hedging (Buehler et al. 2019)

Replace the model delta with a **learned policy** $\delta_t = \pi_\theta(s_t)$ (LSTM) that directly minimises tail risk:

$$\min_{\delta} \ \mathrm{CVaR}_\alpha[-Z(\delta)]$$

- State: moneyness, IV proxy, time-left, lagged position, cost ratio.
- Trained on **7,478 rolling 18-day windows** of TAIEX (1995–2025), strictly pre-backtest.
- Cost-aware objective $\Rightarrow$ rewards *efficient* hedging.

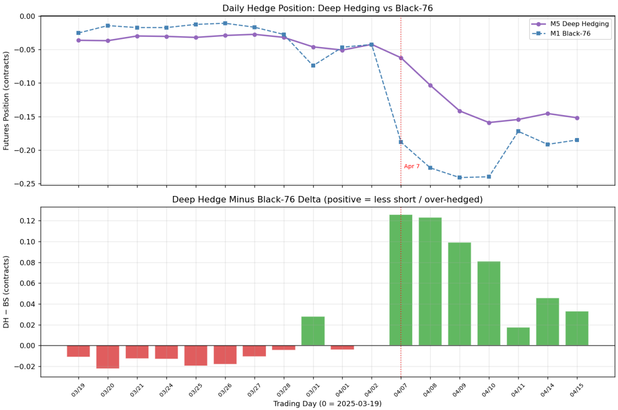
Choosing α — the key design call

I use α=0.95 **(95% Expected Shortfall)** — a principled, conventional tail-risk measure — applied consistently to both training and evaluation, so the objective minimises the expected loss across the **worst 5%** of paths.

Model 5 — A Genuine, Lower-Turnover Hedge

What the network learned (at $\alpha = 0.95$):

- Carries a real hedge: mean $|h| \approx 0.073$ vs. BS 0.097.
- **Front-loads** the hedge in the calm period; ramps smoothly.
- **Under-hedges the crash bottom** — holds 33/46/59% of BS delta on Apr 7–9.
- **Turnover 50% below BS** (0.22 vs. 0.43) $\Rightarrow$ lowest cost, less whipsaw.



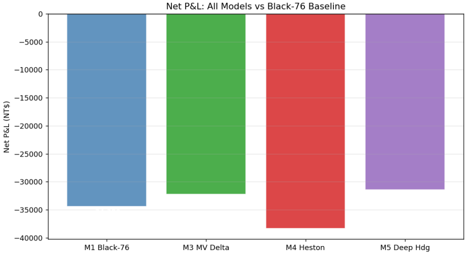
Result — best of all five

Futures P&L −NT\$15,585 (vs. BS −NT\$18,506); Net −NT\$31,434 (+NT\$2,946).

5. Synthesis

Cross-Model Scorecard

Rank	Model	Net P&L (NT$)	vs. Baseline
1	Model 5 — Deep Hedging	−31,434	+2,946
2	Model 3 — MV Delta	−32,244	+2,136
3	Model 1 — Black-76 (baseline)	−34,380	—
4	Model 4 — Heston SV	−38,330	−3,950
5	Model 2b — Sticky-Delta	−39,191	−4,811



- Every model lost money — a 9% OTM put through a jump is structurally hard to hedge.
- The spread from best to worst is **NT$7,757** — model choice clearly matters.

Key Findings

1. Sophistication $\neq$ performance

Heston (most complex classical model) ranked *last* — daily recalibration injected estimation risk that swamped its smile-correction benefit.

2. Robustness beats cleverness with limited data

MV Delta — one stable regression coefficient — gave the most *reliable* classical improvement.

3. The objective *is* the strategy

With a tail-aware risk measure (α=0.95, 95% Expected Shortfall), Deep Hedging learns an efficient, whipsaw-aware hedge that beat the baseline for the **right** reason — turnover control, not abstaining from hedging.

What I Would Do in Production

Risk & objective

- Tail-aware objective (CVaR $\alpha \geq 0.95$), not RMSE delta-replication.
- No-trade bands so rebalancing earns more than it costs.

Model robustness

- Regularise / floor Heston calibration; prefer stable parameters.
- Ensemble the classical delta with the learned policy.

Data & crisis readiness

- Feed **live IV** state, not a backward-looking HV proxy.
- Augment training with simulated **jump** scenarios (tails are scarce).
- Stress-test across α and across crash paths.

Bottom line

Hedge the *tail and the turnover*, not just the textbook delta.

Methodological Rigor (Why the Numbers Are Trustworthy)

- Raw TAIFEX data treated as read-only; settlement prices used exclusively.
- Black-76 with futures-as-forward (shared expiry $\Rightarrow$ no carry fudge).
- IV back-solved same-day; Greeks recomputed from scratch each day.
- Strict no-lookahead — even the LSTM is trained only on pre-t_0 history.
- Attribution reconciles exactly: $\Theta + \Gamma + \mathcal{V} + \Delta + \text{resid} + \text{costs} = \text{Net}$.
- Final settlement cross-checked: SOQ $19{,}548 = 20{,}000 - 452$ (last TXO close).

Thank You

Questions & Discussion

Delta-Neutral Dynamic Hedging | TXO20000P5 | April 2025

Max Chiu

Project repository: `github.com/MaxChiu1122/Delta_Hedging_Strategy`

Appendix — Data Sources

Dataset	Source / note
TXO option chain (Apr 2025)	TAIFEX , 202504 monthly put, regular session
TX futures (Apr 2025)	TAIFEX, `settlement_price`; 03/19 Apr TX = 22,018
CBC interest rates	31–90d CP secondary market ≈ 1.57–1.60%
TAIEX price index	Daily close, 1995–2025 (deep-hedge training)
Final settlement	Official TAIFEX SOQ 202504 = 19,548

Reference: Buehler, Gonon, Teichmann & Wood (2019), *Deep Hedging*, *Quantitative Finance* 19(8), arXiv:1802.03042.